Examiner only

Answer **all** questions.

1. (a) State the principle of conservation of momentum. [2]

(b) Natalie attempts to retrieve a hat of mass 0.40 kg which is resting on the surface of a frozen pond. To do this, she throws a snowball, of mass 0.20 kg, at a horizontal velocity of $9.0\,\mathrm{m\,s^{-1}}$ towards the hat. It hits the hat and rebounds at $2.0\,\mathrm{m\,s^{-1}}$.

(i) Calculate the magnitude of the velocity of the hat after the collision. [3]

(ii) The snowball is in contact with the hat for 0.14 s. Calculate the mean force experienced by the hat in this time. [2]

(c) Natalie believes that, if the snowball stuck to the hat, the velocity of the hat would be greater than the velocity calculated in (b)(i). **Without further calculation**, discuss whether or not Natalie is correct. [3]

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